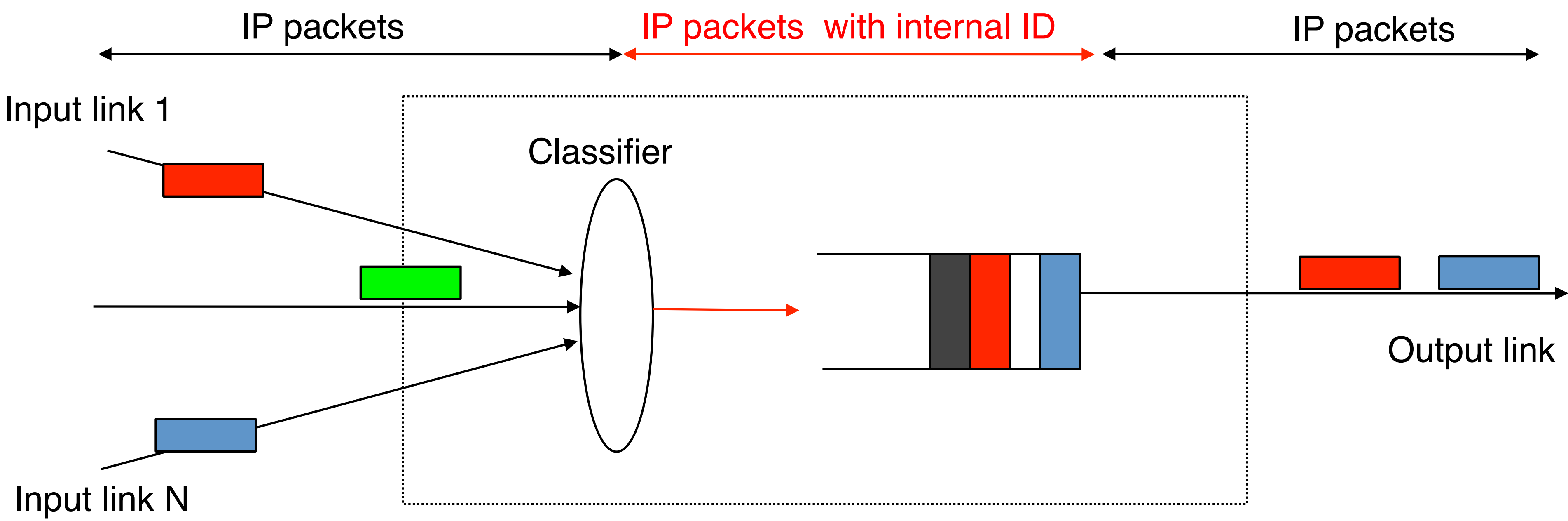
INGI2142

Traffic control techniques

Applications

- What are the different types of QoS-dependent applications and their requirements ?
  - Interactive or not
  - Elastic or not
  - Tolerant or not
  - Adaptive versus non-adaptive
  - Realtime versus streaming
  - Multimedia versus computation

How to identify flows requiring better QoS ?



How would you implement a classifier ?

- Per source classification ?
  - Layer2, layer3, layer4

- Per destination classification ?
  - Layer2, layer3, layer4, layer7

- Par application classification ?

QoS parameters

- Throughput
  - How to unambiguously specify the throughput requirement of an application ?

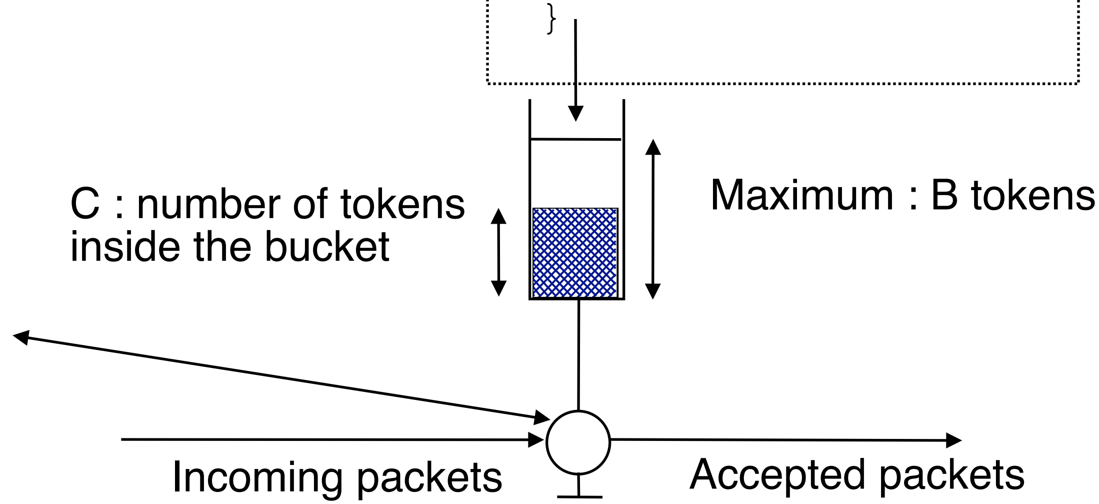
Token bucket

Bucket filling
Initialisation

```
 C=B;
every 1/R second do
 {
  if(C<B)
      C=C+1;
 }
```

Arrival of packet P of length L

```
if (L <= C)
{ /* packet is accepted */
   C=C-L;
}
else
{
  /* packet is discarded */
}
```



Token bucket (2)

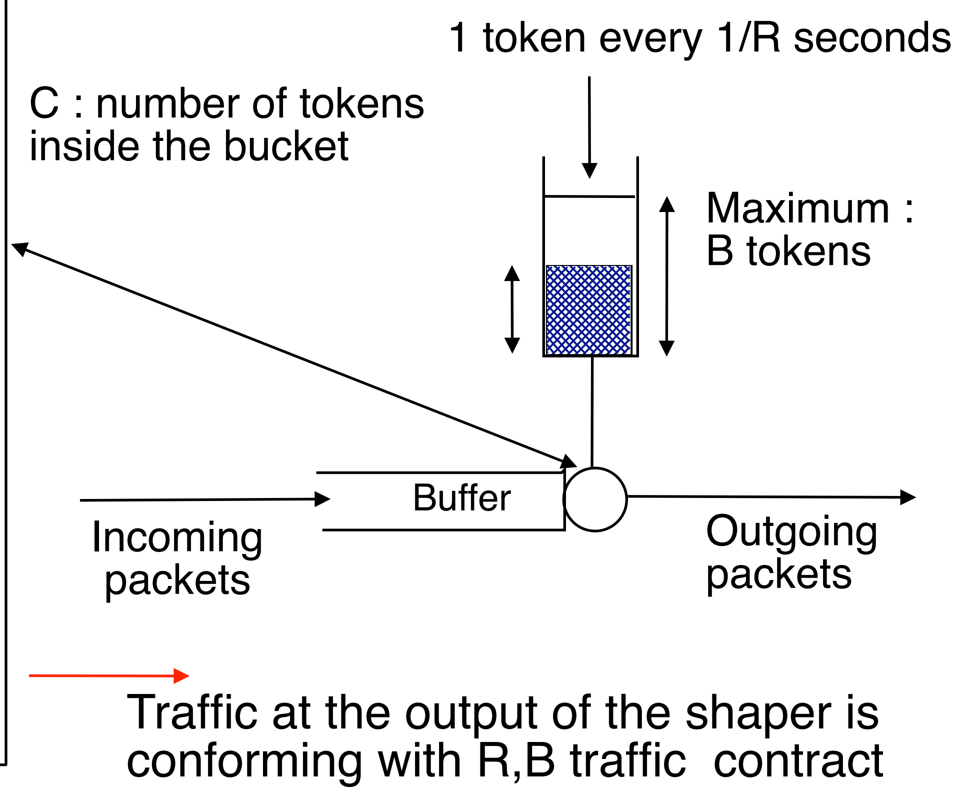
- What is the maximum number of bytes accepted by a token bucket of rate R and burst size B during t seconds ?

Shaping

- Sometimes, we need to modify packets to fit in a token bucket envelope

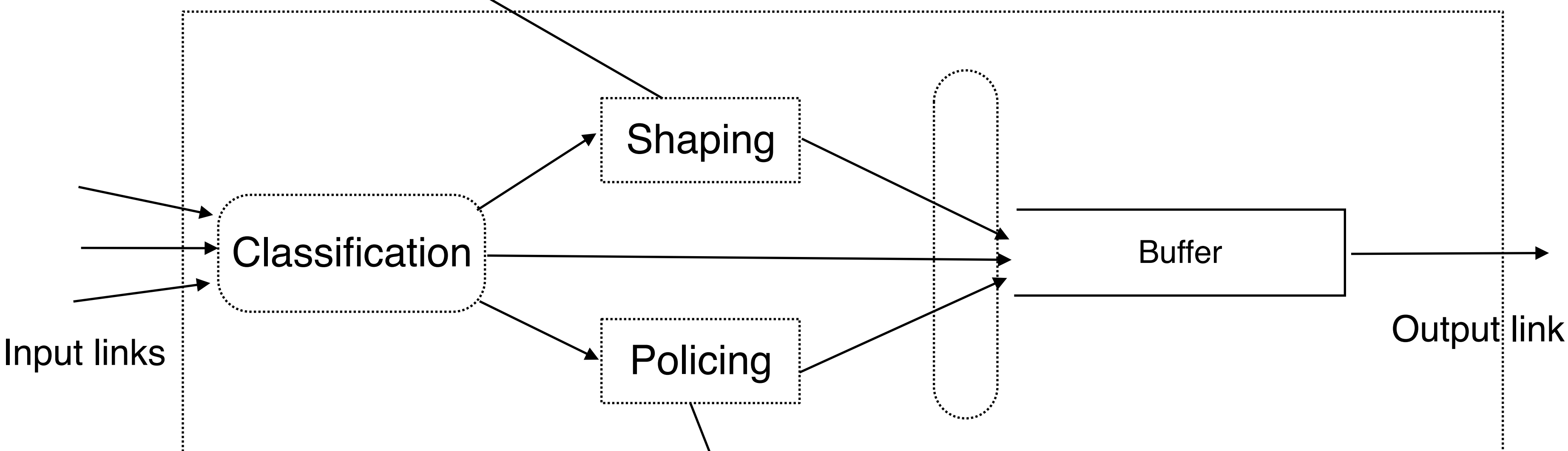
Arrival of packet of size L

```
if (L <= C)
{  /* packet arrived on time */
   C=C-L;
   transmit_packet();
}
else
{/* packet arrived too early
  * delay packet inside buffer
* until it becomes conforming
*/
 while (C<L)
  { /* wait */  }
 /* now C=L and packet is
conforming */
 C=C-L;
 transmit_packet();
}
```



Policing and shaping

Shaper
Enforces a traffic contract by delaying packets so that packets at the output of the shaper conform to the traffic contract



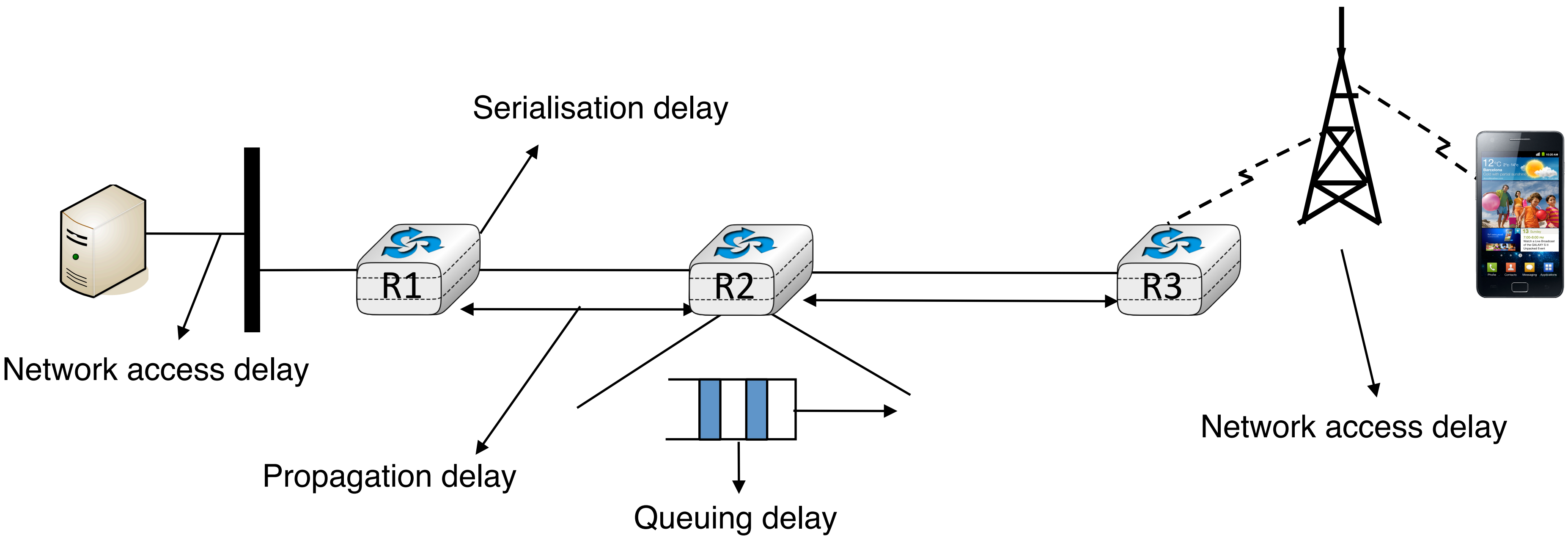
Policer
Enforces a traffic contract and marks/discards the non conforming packets

QoS parameters : delay

- What are the different types of delays experienced by packets in a network ?
  - end-to-end delay
    - one-way or two ways
  - delay jitter

- What are the components of the end-to-end delay ?

Components of end-to-end delay

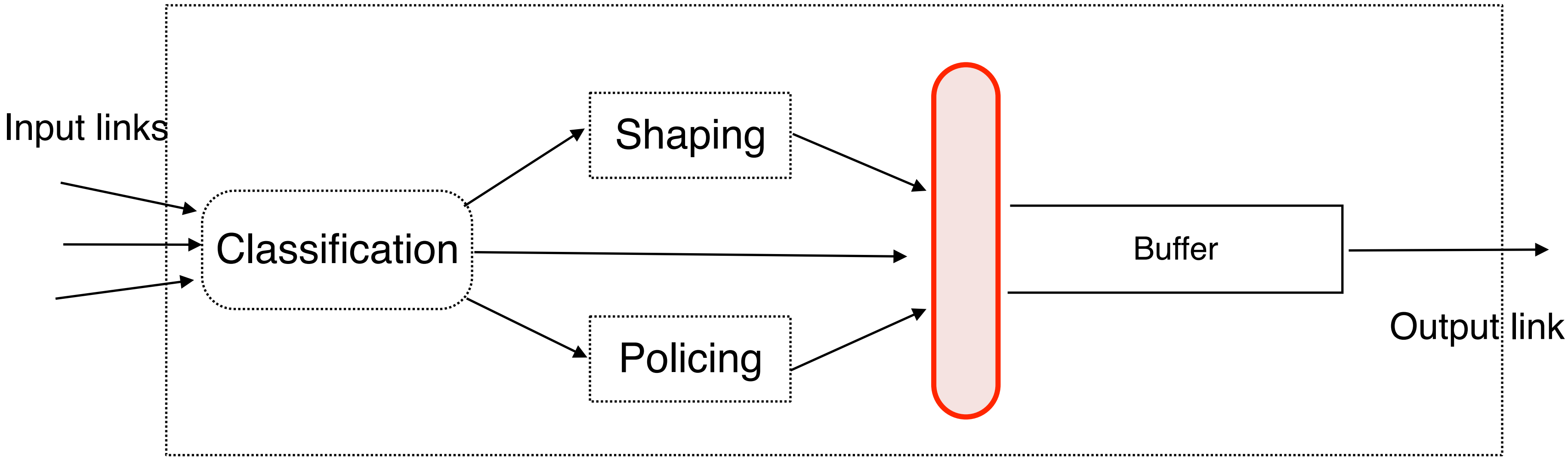


Admission control

- What is admission control ?
  - Is it used in regular IP networks ?

Buffer Acceptance

- How to decide which packets can be accepted inside the buffer ?



Buffer Acceptance

- The two key questions for a buffer acceptance algorithm
  - When should the BA algorithm decide to discard a packet ?
  - Which packet should be discarded ?

Buffer Acceptance Algorithms

- Tail drop

  – Arrival (Packet p) :

```
if isFull(Buffer) then
   discard p;
else
   AddTail(Buffer, p);
```

- What are its advantages and drawbacks ?

Schedulers

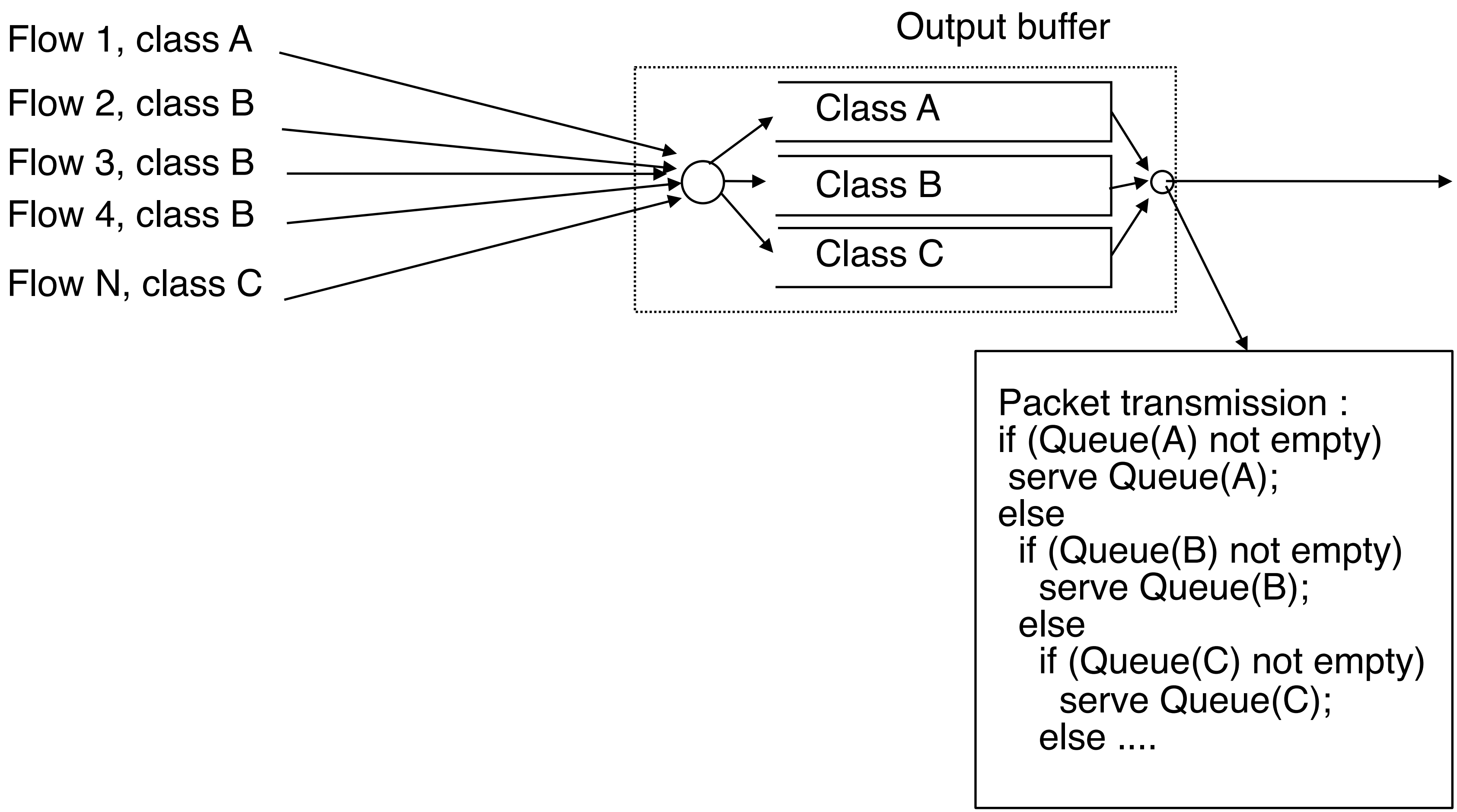
- What are the objectives of scheduling algorithms ?

- What is FIFO scheduling ?
  - How does this scheduler works ?
  - Is it suitable for QoS guarantees ?

Priority scheduling

- Can you explain the operation of the priority scheduler ?
  - What are its advantages and drawbacks ?

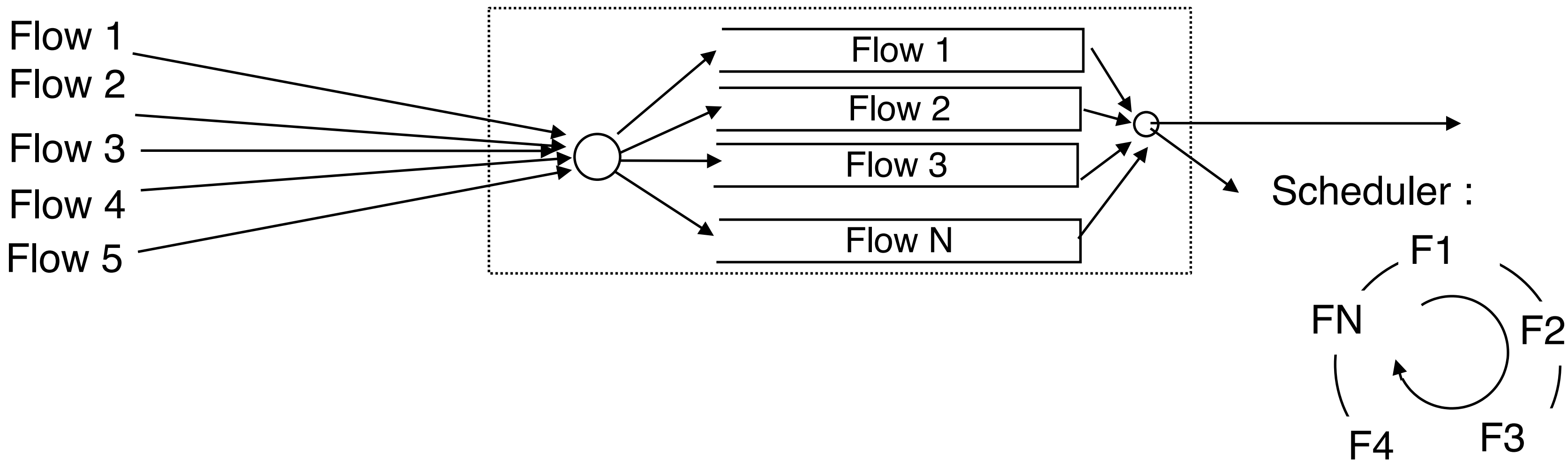
Priority scheduling



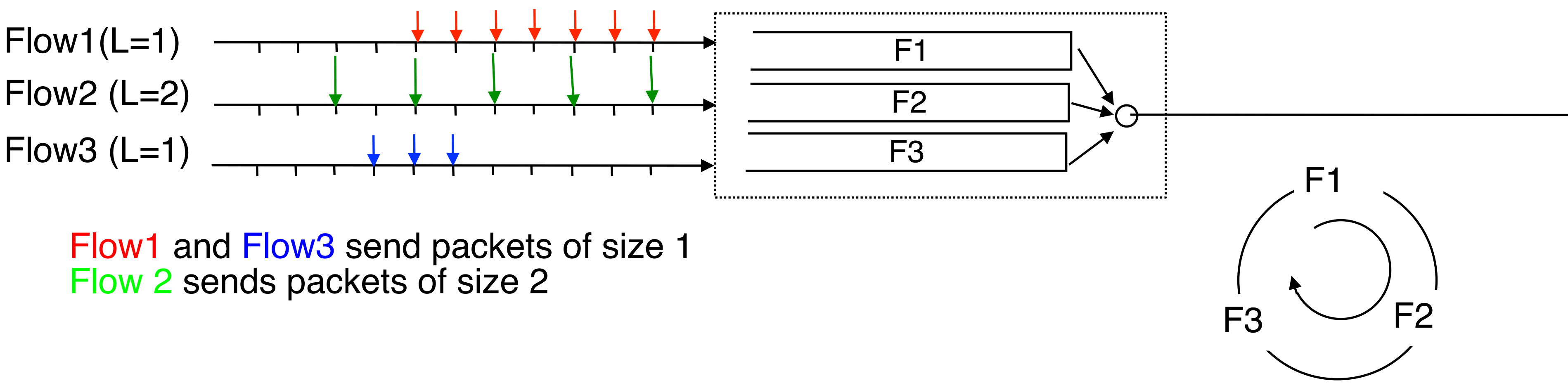
Round-Robin scheduling

- How does this scheduling technique works ?
  - What are its advantages and drawbacks ?

Round-Robin scheduling (2)

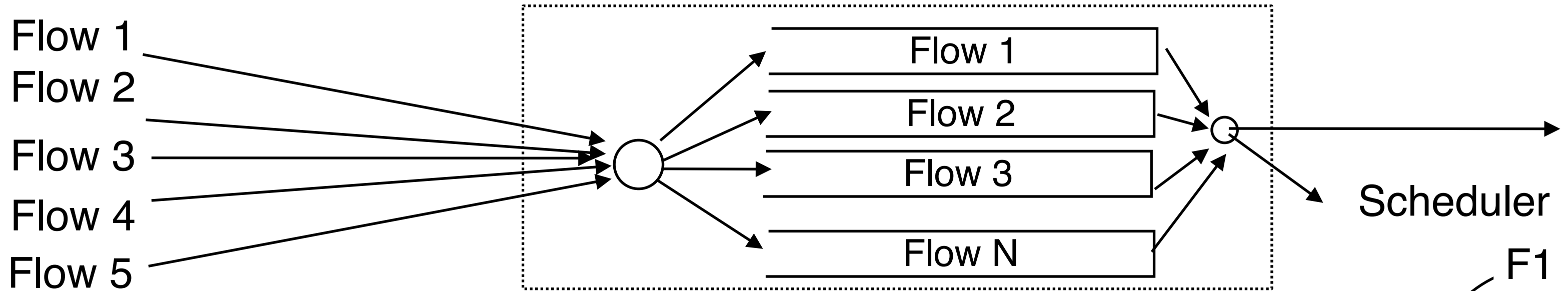


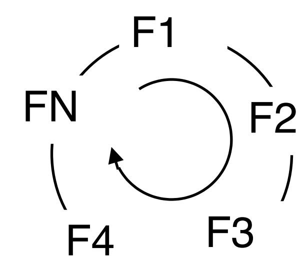
Round-robin : example



Flow1 and Flow3 send packets of size 1
Flow 2 sends packets of size 2

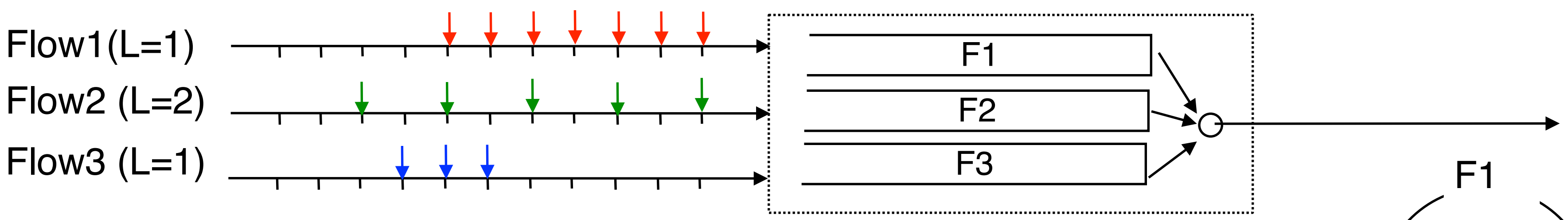
Deficit Round-Robin





- associate counter d[i] to each queue
- increase d[i] by *quantum* every time queue[i] is visited
  - if first_packet of queue[i] larger than d[i]
    { packet stays in queue[i]; }
  - else
    {
    packet is transmitted on output link;
    d[i]=d[i]- packet length;
    if queue[i] is empty { d[i]=0; }
    }

Deficit Round-Robin : example



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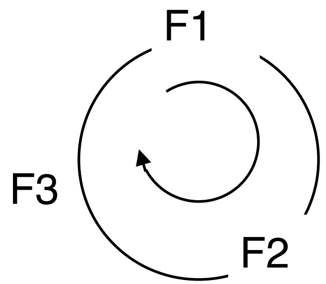
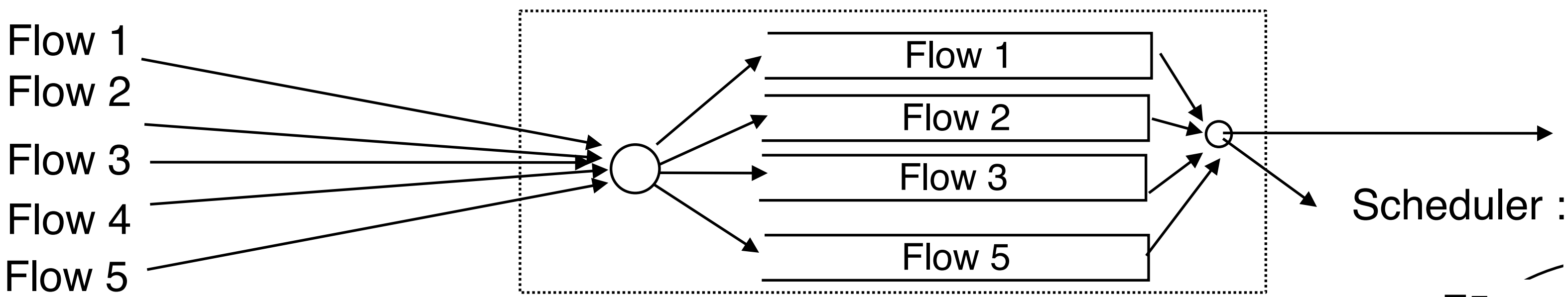


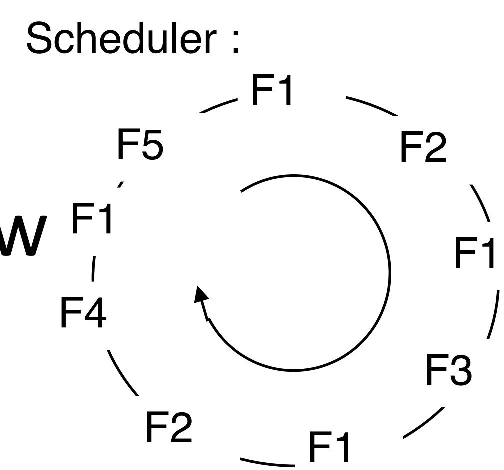
Weighted Round-Robin

- How does this scheduler works ?

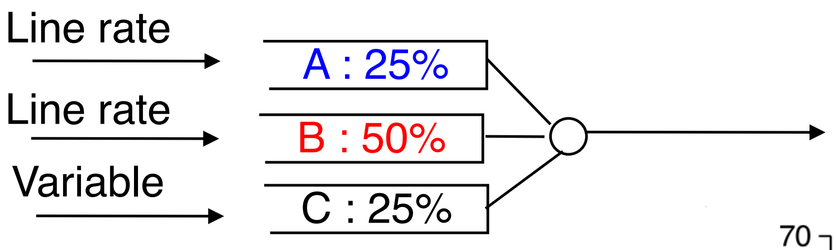
- What are its advantages compared to Round-Robin ?

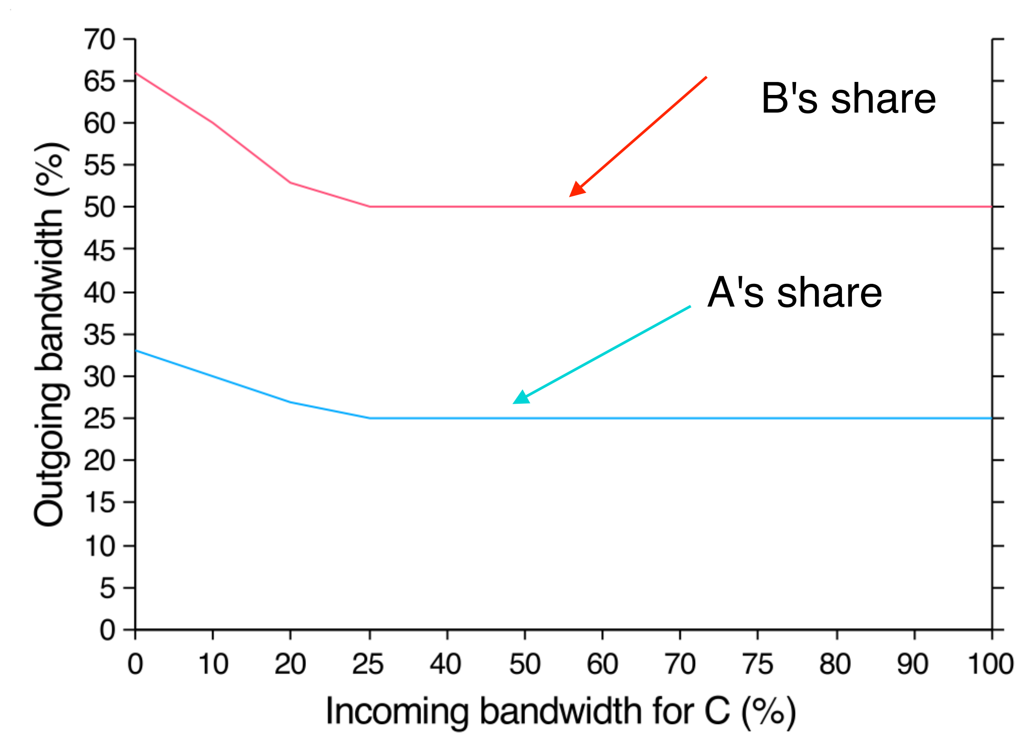


- What is the throughput of each flow

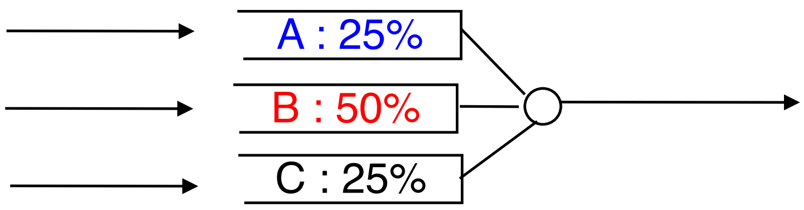


Weighted Round-Robin





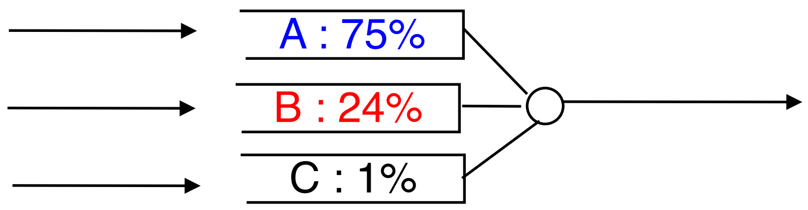
Weighted Round Robin



- Input Traffic pattern
  - A : 20%, regular
  - B: 60%
  - C: 50%

Weighted Round Robin

- What is the impact of a burst of traffic ?



- Input Traffic pattern
  - A : 20%, regular
  - B: 24%
  - C: 100%

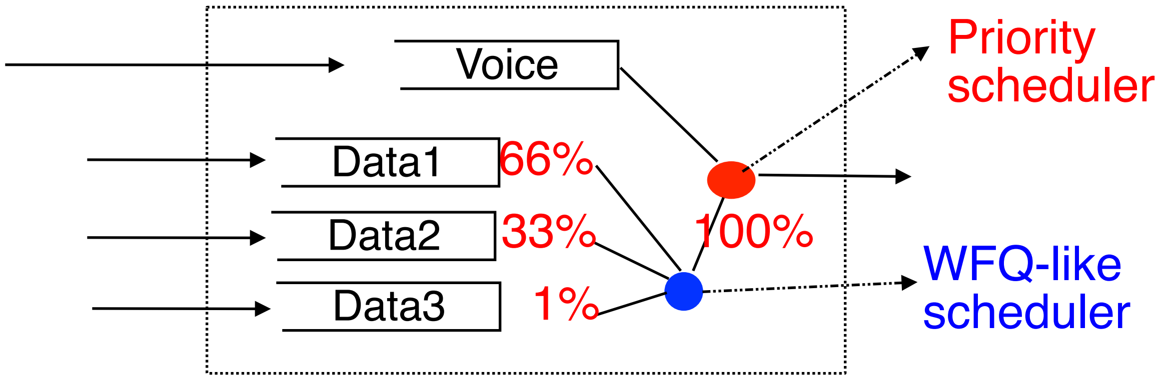
Schedulers and buffer acceptance

- Which buffer acceptance strategy should be used with DRR or WDRR ?
  - Can we identify the flow/queue that is causing congestion ?
  - Which packet should be discarded ?

Weighted Fair Queueing

- What are the key characteristics of this scheduling technique ?
  - What are its benefits compared with DRR ?

Combining schedulers



- Input Traffic pattern
  - 10% voice, regular
  - 30% data1
  - 60% data2
  - 50% data3